

* Clock Sync Example (Christian)

Protocol: ① Client Sends Request

② Server Replies with Timestamp: $t_B = C_B(t)$ (Server B)

③ Client meas. RTT: $\hat{R}$

④ Client Sets Clock: $C_A \leftarrow t_B + \frac{\hat{R}}{2}$ (Precision = $\frac{\hat{R}}{2}$)

Problems: assym. comm. links, Processing delays, Jitter etc.

Berkeley Improves: ① Master Polls all Slaves

② Collects clock values

③ estimate RTTs

④ Removes outliers

⑤ Computes avg. Time (only current poll)

⑥ Sends adjustments to slaves

NTP: Sync Clients to UTC Using Hierarchy of Servers (Strata) (Clever)

- Sync Repeatedly over time
- keep History offset/delay
- Filter noisy statistic
- Can Talk to Mult. Servers and Pick most Reliable uni from Variance



Uni & Laptop Sync

t_1 : LT Req. Time

t_2 : Uni receive time

t_3 : uni Send time

t_4 : LT Receive Time

$$RTT = \underbrace{(t_4 - t_1)}_{\text{Tot. Time}} - \underbrace{(t_3 - t_2)}_{\text{Proc. time}}$$

$$\text{Est. offset} = ((t_2 - t_1) + (t_3 - t_4)) / 2$$

note